

Whittaker Worland

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EDUCATION

University of California, Irvine

Bachelor of Science in Computer Science - Minor in Game Design

Courses: Data Structures & Algorithms, Object-Oriented Programming, Artificial Intelligence, Game Development, UI Design, Cloud Computing, Prompt Engineering, Large Language Models, Natural Language Processing, Human-Computer Interaction

Honors: Dean's List, Scholar-Athlete, 1st Place – UC Irvine Video Game Design Competition

SKILLS

Languages: *Proficient In* - Java, Python, C++, TypeScript *Familiar With* - JavaScript, PHP, R, HTML, CSS, C#, MATLAB

Cloud & Infra: AWS (CDK, DynamoDB, S3, SNS, SQS, EC2, IAM, Lambda, Bedrock, CloudWatch), Docker, Kubernetes

AI & ML: AWS Bedrock, Generative AI, LLMs Integrations, Prompt Engineering, PyTorch, TensorFlow

Frameworks: Node.js, React.js, REST APIs, NumPy, Pandas, Jupyter, Unreal Engine 5, Unity, Blender, Arduino

Core Competencies: Scalable Systems, Infrastructure as Code (IaC), Distributed Systems, Event-Driven Architecture, API Design, Observability, Data Pipelines, Microservices, System Design, CI/CD, Cost Optimization, Agile Development

EXPERIENCE

Amazon - Software Development Engineer - San Diego, CA

March 2024 - Present

Seller Partner Services Org - Seller Partner Financial Foundational Services

- Engineered and launched a new scalable, high-throughput region for a financial data aggregation pipeline handling 100M+ daily requests, migrated legacy region achieving zero-downtime cutover with 100% data integrity and coordinating cross-functional collaboration across 200+ downstream client teams to ensure high availability
- Designed end-to-end AWS infrastructure using Infrastructure as Code (IaC) via AWS CDK, provisioning DynamoDB, S3, SNS, SQS, EC2, and CloudWatch to automate deployments, backfill data, enable fault-tolerant, event-driven architecture
- Achieved cost optimization of ~\$230K/month (~\$2.76M annually) by reducing legacy Ireland region footprint, surpassing compliance goals while enabling scalable growth across future regions
- Served as on-call engineer for production incidents including responses during large-scale events including the October 2025 AWS outage, ensuring rapid recovery with minimal customer impact across millions of active workflows
- Authored internal documentation and best practices for multi-region expansions, establishing repeatable system design patterns adopted by the broader engineering org

Creators Org - Product Advertising API

- Modernized legacy VIP-based architecture by implementing AWS Network Load Balancers, improving system scalability, traffic management, and monitoring visibility across production environments
- Developed local testing support that removed ~90% of deployment friction, empowering engineers to validate code locally and saving our team multiple days per development cycle, which dramatically accelerated developer feature delivery and productivity

Amazon - Software Development Engineer Intern - San Diego, CA

June 2023 - September 2023

Seller Partner Services Org - Seller Partner Financial Foundational Services

- Architected SimBot, an AI-powered automation platform on event-driven architecture (SNS/SQS), cutting away team response times by ~95% (days to minutes) via AWS Bedrock integrations with models like OpenAI, Anthropic, DeepSeek
- Built SimBot's scalable AWS infrastructure using IaC (AWS CDK), DynamoDB, and SQS-driven data pipelines; built SOP-based knowledge bases and applied prompt engineering to optimize LLM behavior from production user feedback
- Designed REST API contracts and data ingestion workflows to integrate SimBot with internal services, enabling cross-functional collaboration with international teams to streamline onboarding, code reviews, and deployments

B.well Inc - Software Engineering Intern - Remote

June 2022 - September 2022

- Integrated customer health data using Epic FHIR APIs, enabling reliable synchronization with MyChart applications and improving data accuracy for thousands of users
- Developed full-stack tools to visualize and manage electronic health records (EHRs), improving usability, data accessibility, and patient engagement across web platforms
- Built, containerized, and deployed services using Docker and Kubernetes, supporting scalable development workflows and consistent environments across local and cloud deployments
- Collaborated within an agile engineering team to design, implement, and test RESTful APIs, strengthening frontend-backend integration through peer code reviews and automated testing
- Applied object-oriented design principles and modern JavaScript best practices to deliver maintainable, production-ready code in a regulated healthcare environment

PROJECTS

Personal & Academic

- Directed a team to build *The Ninth Circle*, a first-person shooter horror roguelike developed in Unreal Engine 5, winning UC Irvine's \$5,000 Video Game Design Competition for outstanding design and gameplay systems
- Designed procedurally-generated worlds and complex game mechanics, demonstrating leadership in team-based creative gaming projects
- Built a calorie tracking app that uses publicly available nutrition data to calculate calorie totals from food weights, providing users with simple meal-logging functionality
- Completed machine learning projects using NumPy and Pandas to analyze datasets and visualize model performance
- Wrote assembly programs to optimize performance and explore low-level memory management and instruction-level parallelism, strengthening understanding of computer architecture and systems efficiency

LEADERSHIP AND TEAMWORK

Leadership

- Team Captain & Scholar-Athlete, Recognized as a Scholar-Athlete for maintaining academic excellence while competing at a Division I level. In high school, led teammates to back-to-back CIF Championships through strong communication, accountability, problem-solving, and strategic coordination
- Guided new team members and a summer intern on complex engineering workflows, reviewing code, clarifying design decisions, and ensuring consistent implementation standards across projects. Fostered a collaborative environment that improved development velocity and onboarding efficiency

Teamwork

- Contributed to multiple team-based software engineering projects in the industry, bringing creative and practical solutions to meet deadlines efficiently. Regularly served as a project lead or code reviewer, ensuring code quality and clear communication across contributors
- Experienced in making high-stakes decisions under time constraints in athletic competition and in collaborative development environments, building strong adaptability and composure under pressure